MSG No. 148.20 $R_I\bar{3}$ [Type IV, trigonal]

Table 1: Wyckoff site: `6a`, site symmetry: `-3..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}\right]$	`[7,8,9,10,11,12]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}\right]$	`[13,14,15,16,17,18]`
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[19,20,21,22,23,24]`
5	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}\right]$	`[25,26,27,28,29,30]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}\right]$	`[31,32,33,34,35,36]`

Table 2: Wyckoff site: `6b`, site symmetry: `-3'..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,22,23,24]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,5,6,19,20,21]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{7}{12}\right]$	`[7,8,9,28,29,30]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{12}\right]$	`[10,11,12,25,26,27]`
5	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{11}{12}\right]$	`[13,14,15,34,35,36]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{5}{12}\right]$	`[16,17,18,31,32,33]`

Table 3: Wyckoff site: `12c`, site symmetry: `3..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3]`
2	$[0,\ 0,\ -z]$	`[4,5,6]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{3}\right]$	`[7,8,9]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{3}-z\right]$	`[10,11,12]`
5	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{2}{3}\right]$	`[13,14,15]`
6	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{2}{3}-z\right]$	`[16,17,18]`
7	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[19,20,21]`
8	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[22,23,24]`
9	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{5}{6}\right]$	`[25,26,27]`
10	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{5}{6}-z\right]$	`[28,29,30]`
11	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{6}\right]$	`[31,32,33]`
12	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{6}-z\right]$	`[34,35,36]`

Table 4: Wyckoff site: `18d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[1,22]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,23]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,24]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[4,19]`
5	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,20]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[6,21]
7	$\left[\frac{1}{6}, \frac{1}{3}, \frac{7}{12}\right]$	[7,28]
8	$\left[\frac{2}{3}, \frac{5}{6}, \frac{7}{12}\right]$	[8,29]
9	$\left[\frac{1}{6}, \frac{5}{6}, \frac{7}{12}\right]$	[9,30]
10	$\left[\frac{1}{6}, \frac{1}{3}, \frac{1}{12}\right]$	[10,25]
11	$\left[\frac{2}{3}, \frac{5}{6}, \frac{1}{12}\right]$	[11,26]
12	$\left[\frac{1}{6}, \frac{5}{6}, \frac{1}{12}\right]$	[12,27]
13	$\left[\frac{5}{6}, \frac{2}{3}, \frac{11}{12}\right]$	[13,34]
14	$\left[\frac{1}{3}, \frac{1}{6}, \frac{11}{12}\right]$	[14,35]
15	$\left[\frac{5}{6}, \frac{1}{6}, \frac{11}{12}\right]$	[15,36]
16	$\left[\frac{5}{6}, \frac{2}{3}, \frac{5}{12}\right]$	[16,31]
17	$\left[\frac{1}{3}, \frac{1}{6}, \frac{5}{12}\right]$	[17,32]
18	$\left[\frac{5}{6}, \frac{1}{6}, \frac{5}{12}\right]$	[18,33]

Table 5: Wyckoff site: 18e, site symmetry: -1

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,5]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,6]
4	$\left[\frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right]$	[7,10]
5	$\left[\frac{2}{3}, \frac{5}{6}, \frac{1}{3}\right]$	[8,11]
6	$\left[\frac{1}{6}, \frac{5}{6}, \frac{1}{3}\right]$	[9,12]
7	$\left[\frac{5}{6}, \frac{2}{3}, \frac{2}{3}\right]$	[13,16]
8	$\left[\frac{1}{3}, \frac{1}{6}, \frac{2}{3}\right]$	[14,17]
9	$\left[\frac{5}{6}, \frac{1}{6}, \frac{2}{3}\right]$	[15,18]
10	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[19,22]
11	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[20,23]
12	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[21,24]
13	$\left[\frac{1}{6}, \frac{1}{3}, \frac{5}{6}\right]$	[25,28]
14	$\left[\frac{2}{3}, \frac{5}{6}, \frac{5}{6}\right]$	[26,29]
15	$\left[\frac{1}{6}, \frac{5}{6}, \frac{5}{6}\right]$	[27,30]
16	$\left[\frac{5}{6}, \frac{2}{3}, \frac{1}{6}\right]$	[31,34]
17	$\left[\frac{1}{3}, \frac{1}{6}, \frac{1}{6}\right]$	[32,35]
18	$\left[\frac{5}{6}, \frac{1}{6}, \frac{1}{6}\right]$	[33,36]

Table 6: Wyckoff site: 36f, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-y, x-y, z]$	[2]
3	$[-x+y, -x, z]$	[3]
4	$[-x, -y, -z]$	[4]
5	$[y, -x+y, -z]$	[5]

continued ...

Table 6

No.	position	mapping
6	$[x-y,\ x,\ -z]$	[6]
7	$\left[x+\frac{2}{3},\ y+\frac{1}{3},\ z+\frac{1}{3}\right]$	[7]
8	$\left[\frac{2}{3}-y,\ x-y+\frac{1}{3},\ z+\frac{1}{3}\right]$	[8]
9	$\left[-x+y+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{1}{3}\right]$	[9]
10	$\left[\frac{2}{3}-x,\ \frac{1}{3}-y,\ \frac{1}{3}-z\right]$	[10]
11	$\left[y+\frac{2}{3},\ -x+y+\frac{1}{3},\ \frac{1}{3}-z\right]$	[11]
12	$\left[x-y+\frac{2}{3},\ x+\frac{1}{3},\ \frac{1}{3}-z\right]$	[12]
13	$\left[x+\frac{1}{3},\ y+\frac{2}{3},\ z+\frac{2}{3}\right]$	[13]
14	$\left[\frac{1}{3}-y,\ x-y+\frac{2}{3},\ z+\frac{2}{3}\right]$	[14]
15	$\left[-x+y+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{2}{3}\right]$	[15]
16	$\left[\frac{1}{3}-x,\ \frac{2}{3}-y,\ \frac{2}{3}-z\right]$	[16]
17	$\left[y+\frac{1}{3},\ -x+y+\frac{2}{3},\ \frac{2}{3}-z\right]$	[17]
18	$\left[x-y+\frac{1}{3},\ x+\frac{2}{3},\ \frac{2}{3}-z\right]$	[18]
19	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[19]
20	$\left[-y,\ x-y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[-x+y,\ -x,\ z+\frac{1}{2}\right]$	[21]
22	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	[22]
23	$\left[y,\ -x+y,\ \frac{1}{2}-z\right]$	[23]
24	$\left[x-y,\ x,\ \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{2}{3},\ y+\frac{1}{3},\ z+\frac{5}{6}\right]$	[25]
26	$\left[\frac{2}{3}-y,\ x-y+\frac{1}{3},\ z+\frac{5}{6}\right]$	[26]
27	$\left[-x+y+\frac{2}{3},\ \frac{1}{3}-x,\ z+\frac{5}{6}\right]$	[27]
28	$\left[\frac{2}{3}-x,\ \frac{1}{3}-y,\ \frac{5}{6}-z\right]$	[28]
29	$\left[y+\frac{2}{3},\ -x+y+\frac{1}{3},\ \frac{5}{6}-z\right]$	[29]
30	$\left[x-y+\frac{2}{3},\ x+\frac{1}{3},\ \frac{5}{6}-z\right]$	[30]
31	$\left[x+\frac{1}{3},\ y+\frac{2}{3},\ z+\frac{1}{6}\right]$	[31]
32	$\left[\frac{1}{3}-y,\ x-y+\frac{2}{3},\ z+\frac{1}{6}\right]$	[32]
33	$\left[-x+y+\frac{1}{3},\ \frac{2}{3}-x,\ z+\frac{1}{6}\right]$	[33]
34	$\left[\frac{1}{3}-x,\ \frac{2}{3}-y,\ \frac{1}{6}-z\right]$	[34]
35	$\left[y+\frac{1}{3},\ -x+y+\frac{2}{3},\ \frac{1}{6}-z\right]$	[35]
36	$\left[x-y+\frac{1}{3},\ x+\frac{2}{3},\ \frac{1}{6}-z\right]$	[36]